



S2 – Mobile Bed and Flow Visualisation Tanks

The S2 – Mobile Bed and Flow Visualisation Tanks are used in two principal fields of study. The first involves detailed investigation of mobile bed situations. These may be in relation to watercourses or civil engineering structures.

The second field involves two dimensional flow visualisation. This may be undertaken using the Ahlborn (c.1902) dust indicator technique or by any other suitable method of flow visualisation.

PRODUCT CODE: S2

Categories: Educational, S Series - Applied Hydraulics & Hydrology, S Series - Hydraulics & Hydrology, S2 - Mobile Bed And Flow Visualisation Tanks

Tags: Ahlborn Technique , Bridge Piers , Civil Engineering , Civil Engineering Structures , Curved Flow , Eddy Shedding , Erosion And Deposition , Fluid Flow , Free Jets , Gated Weir , Groynes And Projections , Hydraulic Jump , Hydraulic Model Studies , Hydraulics & Hydrology , Meandering Rivers , Meandering Water Courses , Mobile Bed And Flow Visualisation Tanks , Oscillating Point Of Separation , Ripple Formation , S Series , S2 , Suction Demonstration , Symmetrical Aerofoil , Tee Junctions In Irrigation Systems , Two Dimensional Flow Visualisation , Watercourses

DESCRIPTION

Practical demonstration and visualisation are essential elements of fluid flow study.

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The tank is moulded from self coloured Oxford Blue (BS0105) glass fibre reinforced plastic with steel reinforcements to provide rigidity. It is manufactured in three sections; the inlet tank, the working section and the discharge reservoir tank. The

sections are joined by flanged connections and despatched from the factory as a complete assembly. A drop-tight adjustable overshoot weir with upstream sand trap is accommodated within the discharge tank. The inlet tank features a perforated baffle plate which spreads the flow evenly across the width of the table.

A removable glass sheet coloured blue on one side and white on the other, is provided to cover the sand bed when flow visualisation experiments are in progress. A pair of adjustable aluminium instrument rails are fitted to the top of the tank. These extend over the full length of the working section and one rail carries a positioning scale.

The depth gauge supplied is used to measure the water level and to map the contours of the sand bed produced during exercises.

It is provided with a stainless steel hook and point and incorporates a Vernier scale enabling levels to be determined accurately.

The gauge is designed for mounting on the instrument carrier assembly which can be positioned over any point of the working area. The main carrier (longitudinal traverse) is provided with a locking device and cursor to operate in conjunction with the instrument rail scale.

The sub-carriage (transverse traverse) operates on rails provided by the main carriage. There is again a transverse scale and locking device. The correct use of the apparatus is described in a comprehensive manual supplied with the tank.

Water flow is provided by a centrifugal pump made of materials which are selected for their corrosion resistance as is the pipework which includes an electrically operated flow regulating valve. A GRP moulding carries the pump, control valve, rigid pipework and 'in-line' flowmeter.

The assembly is connected to the tank by flexible hoses. The motor starter and digital meter readout are mounted in a table top cabinet supplied complete with the required flexible cables.

The tank is available with optional working lengths:

Model S2-2M: working length 2m

Model S2-4M: working length 4m

S2 – Mobile Bed and Flow Visualisation Tanks

TECHNICAL SPECIFICATIONS

Working area: 2 m x 610mm or

4 m x 610mm

Max water depth: 120mm

Thickness of sand bed: 60mm

Flow range: 0-3.6 litres/sec

Sump capacity: 300 litres

Accuracy of flow metering: $\pm 1.5\%$ of full scale deflection



FEATURES & BENEFITS

- **Self-contained recirculating water tank:** Ideal for flow visualization and mobile bed studies, providing a controlled and efficient learning environment.
- **Durable construction:** Made from glass-reinforced plastic with non-corroding materials, ensuring long-lasting use even in water-based experiments.
- **Spacious working section:** With dimensions of 2m (or 4m) x 610mm and a flow range of 0-3.5 liters/sec, it accommodates a wide variety of experiments.
- **Versatile models and accessories:** Comes with 15 models and accessories, plus a colored glass sheet for easy transition between mobile bed and flow visualization modes.
- **Portable control console:** Housed in a portable console with water-safe connectors and flexible cable, offering ease of use and setup for a vast array of experiments.

This setup provides flexibility, durability, and a comprehensive toolkit for fluid mechanics studies.

Experimental content

Mobile bed experiments

- Two-dimensional flow patterns by the Ahlborn technique
 - Converging flow in a curved entry
 - Diverging flow from a nozzle
 - Flow around isolated bodies
 - Circular cylinder
 - Rectangular block
 - Tee shapes
 - Symmetrical aerofoil parallel to the flow
 - Symmetrical aerofoil at incidence
 - Diamond shape with and without suction
 - Arrays of cylinders
 - Free jets
 - A turbulent jet
 - A laminar jet
 - Confined zones of separated flow
 - Flow past a gate-slot
 - Elongated tee and step
 - Simulation of a hydraulic jump
 - Protection for a spillway toe
- Velocity distribution in duct flow
 - Flow in a smooth-walled pipe
 - Flow past a rough boundary
- Hydraulic analogy to compressible flow
- Three-dimensional flow patterns by the Ahlborn technique
 - Influence of normal forces on boundary layer flows
 - Decelerating flow near a corner
 - Boundary layer flow in a curved path
 - Simulations of practical three-dimensional flow patterns

Bridge piers and piles
Spurs (or groynes) and wall discontinuities
Tee junctions in irrigation systems
Meandering rivers

- Characteristics of meandering water courses
- Experimental investigation of erosion and deposition
- Boundary layer suction demonstration
- Boundary layer flows
- Experiments with loose boundaries
 - Bed movement near a circular cylinder
 - Alleviation of erosion around a cylinder
 - Bed movement near non-circular bridge piers
 - Bed movement near groynes and projections
 - Ripple formation in fine sand
- Hydraulic model studies in an Ahlborn tank
 - Operation of a gated weir
 - Cause of damage to an irrigation drop structure
 - Curved flow in a channel with a bend
 - Forces on moored ships
 - Circulation in a lock
- Unsteady flow patterns
 - Unsteady boundary layer growth of a cylinder
 - Eddy generation of an oscillating point of separation
 - Flow patterns near a separation point
 - Unsteady flows near stationary boundaries
 - Eddy shedding from a cylinder
 - Excitation of lateral standing waves in a flow channel
 - Flow-induced vibration of boundaries
 - Vibration of cylinders
 - Lateral vibration of a box

S2-Converging-Flow

S2-Diverging-Flow



S2-Turbulent-Jet-Diffusion

S2 Video

Ordering specification

- A self contained recirculating water tank for flow visualisation and mobile bed studies
- The tank is manufactured from glass reinforced plastic and all components in contact with water are of non-corroding materials
- The working section has minimum dimensions of 2m (4m) x 610mm and the flow range is 0-3.5 litres/sec
- Fifteen models and accessories are included as standard and a sheet of coloured glass allows rapid changeover from mobile bed to flow visualisation mode
- All controls are housed in a portable console which includes a flexible cable and water safe connectors

Supplied Accessories

A wide range of models and accessories is supplied as standard with the Mobile Bed and Flow Visualisation Tank

- Hook and point gauge
- Plasticine
- Water soluble paint (white & black)
- Hypodermic tube & flexible tube
- Rectangular bridge piers (x2)
- Cylindrical bridge piers (x2)
- Round end bridge piers (x2)
- Special bridge piers (x2)
- Aerofoil
- Gate grooves (x2)
- Angles (x6)
- Tees (x12)
- Cylinders
- Set of tank strips/baffles (2x 225mm, 2x 300mm, 2x 600mm, 2x 900mm)
- Side wall meanders (x2)
- Pendulum support frame
- Small pendulum
- Large pendulum
- Swing box
- 90° angle walls (x2)
- Clear acrylic cylinder (perspex)
- Irrotational bend
- Syphon tube
- Bellmouth entries (x2)
- Retaining block
- Swing box weight
- Cork floats (x5)

Requirements

Electricity supply:

S2-2M-A: 220/240V/1ph/50Hz

S2-2M-B: 120V/1ph/60Hz

S2-4M-A: 220/240/1ph/50Hz

S2-4M-B: 120V/1ph/50Hz

Cold water

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

S2-2M

Volume: 3.70m³

Gross Weight: 590kg

S2-4M



Volume: 5.70m³
Gross Weight: 750kg

Dimensions

S2-2M

Length: 3.70m
Width: 0.71m
Height: 2.0m

S2-4M

Length: 5.70m
Width: 0.71m
Height: 2.0m

Order Codes

- S2-2M-A
- S2-2M-B
- S2-4M-A
- S2-4M-B

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